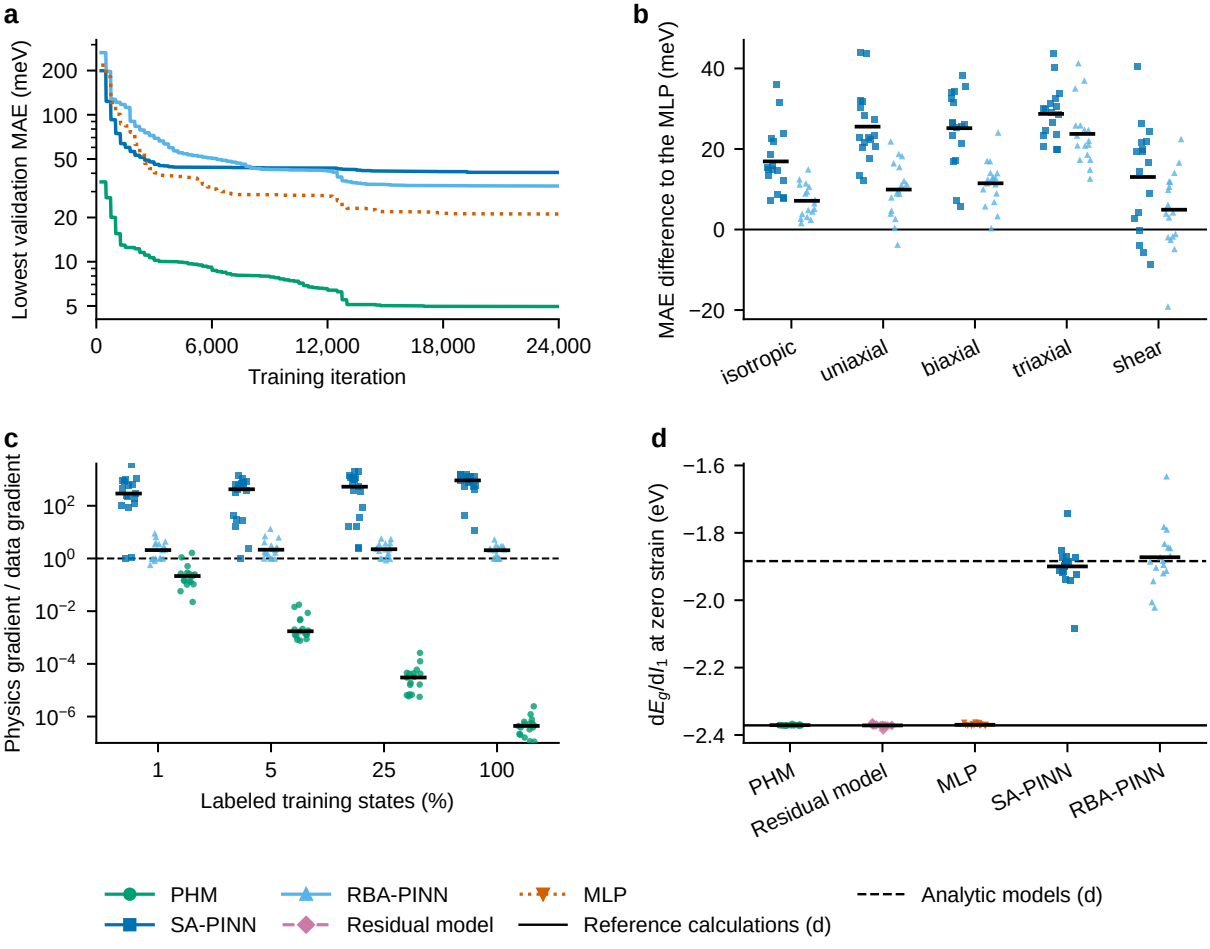
a
Lowest validation MAE (meV)
200
100
50
20
10
5
0
6,000
12,000
18,000
24,000
Training iteration
b
MAE difference to the MLP (meV)
40
20
0
−20
isotropic
uniaxial
biaxial
triaxial
shear
c
Physics gradient / data gradient
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
1
5
25
100
Labeled training states (%)
d
dE_g/dI_1 at zero strain (eV)
−1.6
−1.8
−2.0
−2.2
−2.4
PHM
Residual model
MLP
SA-PINN
RBA-PINN
PHM
SA-PINN
RBA-PINN
Residual model
MLP
Reference calculations (d)
Analytic models (d)